

RETHINKING BIOLOGY

A Journey from Molecules to Information

ARTICLE 4

THE QUIET REVOLUTION CALLED QUANTUM BIOLOGY

When the strange laws of the quantum world began appearing inside living systems

Life is warm, wet and noisy—everything quantum systems are not supposed to survive in. Yet, nature has found ways to exploit quantum phenomena to perform essential biological functions.



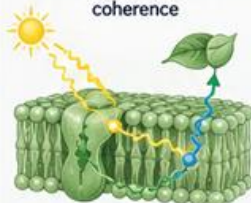
"Nature does not need an engineer to use quantum effects. Evolution preserves arrangements that work."

QUANTUM PHENOMENA IN BIOLOGICAL SYSTEMS

Growing evidence shows that quantum effects can influence key biological processes.

1. PHOTOSYNTHESIS

Light guided with quantum coherence

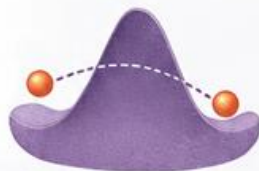


Excitation energy moves through complex molecular networks with remarkable efficiency. Quantum coherence and interference may help energy find the best route to the reaction center.

Nature's antennas may use quantum physics to harvest sunlight with extraordinary efficiency.

2. ENZYME REACTIONS

Tunnelling through energy barriers

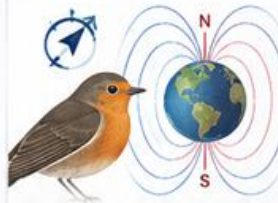


Quantum tunnelling allows electrons or protons to cross energy barriers that would be insurmountable in classical physics—accelerating many biochemical reactions.

Tiny quantum shortcuts can make the chemistry of life possible.

3. MAGNETORECEPTION

Sensing Earth's magnetic field



Quantum spin dynamics in radical pairs may be sensitive enough to detect Earth's weak magnetic field—helping birds and other organisms navigate.

Quantum spins could be nature's compass.

4. OLFACTION

Smell beyond shape



Molecular vibrations and quantum properties may help explain how we distinguish smells that look similar but smell different.

Smell may be a quantum information phenomenon.

WHAT DOES "QUANTUM BIOLOGY" MEAN?

Level 1



Quantum mechanics underlies all chemistry.

Level 2



Quantum effects influence molecular reactions.

Level 3



Quantum phenomena contribute to biological function.

Level 4



Could biological systems actively exploit quantum information processing?

The first three levels are being investigated scientifically.

Level 4 remains speculative and open to research.

QUANTUM BIOLOGY IS NOT...



Not proof that cells are quantum computers.



Not evidence that consciousness is quantum.



Not proof of telepathy, wormholes or universal consciousness.



Not a replacement for chemistry or classical biology.

WHY IT MATTERS



Inspires new technologies by learning from nature's quantum strategies.



Pushes biology to explore deeper layers of information and organization.



Bridges physics, chemistry and biology in a new interdisciplinary frontier.

THE BIG QUESTION



If living systems can exploit quantum phenomena, how might these principles shape the way biological systems represent, transmit and process information?

This is the bridge to our next step:
From Chemistry to Information Processing.

THE TAKE-HOME MESSAGE

Quantum biology does not claim that life is mysterious or magical. It simply reveals that the quantum world—strange as it is—can quietly play a role in the elegance of life.



Quantum Physics
The rules of the universe



Quantum Chemistry
Molecules and reactions



Quantum Biology
Quantum effects in living systems



Biological Information
A deeper, richer form of information



Beyond...
Exciting questions await

“The universe is quantum. Life is information. The journey of science is to understand how they meet.”

There was a time when the words **quantum mechanics** and **biology** seemed to belong to two different intellectual worlds.

Quantum mechanics was the science of atoms, electrons and photons. It dealt with phenomena that appeared strange, counterintuitive and often impossibly small. Biology, on the other hand, was the science of living organisms—cells, genes, proteins, plants, animals and ecosystems.

The two worlds seemed separated not only by subject matter but also by conditions.

Quantum systems are often studied under carefully controlled laboratory conditions. Biological systems operate in warm, wet and extraordinarily noisy environments. A biological cell contains thousands of different molecular species, all moving, colliding and interacting with one another. It is difficult to imagine a less hospitable environment for the delicate quantum phenomena that physicists normally associate with isolated systems.

And yet, over the last few decades, a remarkable possibility has emerged.

Perhaps life does not merely obey quantum mechanics. Perhaps, in some circumstances, life makes use of it.

This does not mean that a cell is a miniature quantum computer. It does not mean that every biological phenomenon is mysterious or inexplicable. And it certainly does not establish that consciousness is a quantum phenomenon.

The claim is much more modest—and, for that very reason, perhaps more scientifically interesting.

Some biological processes appear to involve phenomena that are fundamentally quantum mechanical.

Photosynthesis provides one example. Enzyme reactions provide another. The ability of some organisms to detect magnetic fields has generated another. Even the question of how we perceive smells has inspired proposals involving quantum effects.

These examples force us to reconsider a familiar assumption.

Perhaps the relationship between physics and biology is not simply that physics provides the rules and biology follows them.

Perhaps evolution has discovered ways of **exploiting some of the possibilities offered by quantum physics**.

That is the quiet revolution called quantum biology.

The physics hiding underneath biology

At first glance, there is nothing surprising about saying that biology is quantum mechanical.

Everything around us is ultimately made of atoms. Atoms are governed by quantum mechanics. Therefore, in a fundamental sense, everything biological is quantum mechanical.

But this statement is almost too easy.

The interesting scientific question is not whether quantum mechanics exists inside biological organisms. Of course it does.

The deeper question is:

Can specifically quantum phenomena have a functional role in living systems?

There is an important distinction here.

Consider a protein.

The chemical bonds holding the protein together are ultimately quantum mechanical. The electronic structure of its atoms is quantum mechanical. The interactions between its constituent atoms depend upon quantum mechanics.

But we would not normally say that the protein is performing a quantum computation merely because quantum mechanics provides the foundation for its chemistry.

For quantum biology to become a meaningful concept, we need something more.

We need situations in which phenomena such as **superposition, tunnelling, coherence or quantum spin dynamics** may influence biological function in a way that cannot simply be treated as an irrelevant microscopic detail.

That is where the subject becomes fascinating.

My earlier work was motivated in part by precisely this broader question: whether quantum phenomena might play a larger role in biological organization and information processing than is usually assumed. My 2019 review explicitly examined the possibility of quantum properties in complex biological molecules and considered whether such phenomena might contribute to communication within cells.

But before going further, it is useful to separate what is relatively well established from what remains speculative.

Quantum biology is not one single theory.

It is a growing interdisciplinary field containing experimentally supported quantum effects, active areas of investigation, competing explanations and much more speculative proposals.

That distinction will be important throughout this article.

A leaf behaving like a quantum system

Perhaps the most beautiful place to begin is a green leaf.

We normally think of photosynthesis as a biological process so familiar that it has almost lost its sense of wonder.

A plant captures sunlight.

Light energy is converted into chemical energy.

That chemical energy ultimately supports the growth and metabolism of the organism.

But inside the photosynthetic machinery, something remarkable is happening at the molecular level.

When light is absorbed, energy must be transferred through a complicated network of molecules toward the chemical reaction centres where it can be used.

The challenge is not trivial.

The molecular environment is complex. Energy has multiple possible pathways through which it can travel. Yet biological systems have evolved highly efficient molecular architectures for transferring and utilizing that energy.

Researchers have investigated whether quantum coherence and related quantum effects contribute to this process.

The significance of the question is profound.

Imagine that energy moving through a molecular system were simply like a ball rolling randomly through a complicated maze.

The ball would encounter many possible routes. Some would be inefficient. Some would lead nowhere useful.

But quantum mechanics provides a different description.

At the quantum level, possibilities can be represented collectively before a particular outcome is observed. Quantum amplitudes associated with different pathways can interfere with one another.

The biological system therefore does not necessarily have to behave like a collection of tiny classical particles wandering independently through space.

Under suitable conditions, the quantum description of the system can involve relationships between multiple possible pathways.

This does not mean that a leaf is consciously calculating where a photon should go.

It means something more subtle.

The physical rules governing the transfer of energy can retain quantum character long enough to influence a biological process.

That alone changes our intuition about the living cell.

For much of modern biology, we have become accustomed to thinking in terms of molecules behaving like tiny mechanical components. A molecule binds to another molecule. A reaction takes place. A product is formed. A signal is transmitted.

Quantum biology reminds us that underneath this apparently mechanical description is a deeper physical layer.

At that level, molecules are not simply little billiard balls.

They are quantum systems.

And sometimes the distinction matters.

The strange shortcut through a molecular barrier

There is another quantum phenomenon that provides an even more intuitive example.

It is called **quantum tunnelling**.

Imagine a particle approaching a hill.

In the classical world, if the particle does not have enough energy to climb the hill, it cannot get to the other side. It must remain on its original side.

Quantum mechanics changes this picture.

At the quantum scale, a particle has a probability of being detected on the other side of a barrier even when, according to classical mechanics, it does not possess enough energy to cross that barrier in the ordinary way.

It is as though the particle has found a shortcut through the hill.

This is quantum tunnelling.

It sounds like an exotic laboratory phenomenon.

But quantum tunnelling can play a role in chemistry.

Biological chemistry involves enormous numbers of reactions in which electrons or hydrogen nuclei move between molecular sites. In some circumstances, quantum tunnelling can influence the rates of such reactions.

This is particularly interesting for enzymes.

Enzymes are often described as biological catalysts. They accelerate chemical reactions by providing a favourable molecular environment in which reactions can occur.

The classical picture is already remarkable.

But quantum mechanics adds another layer.

Instead of imagining every reacting particle simply climbing over a classical energy barrier, we may sometimes need to consider the possibility that quantum tunnelling contributes to the reaction.

This is a good example of what quantum biology really means.

We do not have to imagine a cell running an abstract quantum algorithm.

We do not have to invoke consciousness.

We do not have to abandon chemistry.

We simply recognize that **the microscopic quantum behaviour of particles can sometimes influence the macroscopic function of a biological molecule.**

This is an important conceptual lesson.

Quantum biology is not necessarily about replacing biology with physics.

It is about discovering where the quantum description adds something biologically meaningful.

When an animal finds its way using Earth's magnetic field

Now consider something completely different.

A migratory bird begins a journey.

It may travel thousands of kilometres.

It does not have a magnetic compass manufactured in a laboratory.

It does not carry a map.

And yet many animals possess the remarkable ability to sense Earth's magnetic field and use it as part of their navigation.

How?

Magnetoreception has been studied from many perspectives, and one influential class of hypotheses involves **radical-pair chemistry**.

Here the story becomes particularly interesting because quantum spin dynamics may influence the chemical fate of molecular pairs.

The Earth's magnetic field is extremely weak compared with many familiar magnetic fields. Yet biological systems may possess molecular mechanisms sensitive enough to respond to it.

The possibility that quantum spin processes contribute to magnetoreception is one of the reasons quantum biology has attracted such attention.

The example also illustrates something important about biological evolution.

Evolution does not care whether a mechanism looks elegant from the perspective of classical physics.

If a physical effect exists and provides a selective advantage, biological systems can potentially exploit it.

Evolution has had billions of years to experiment.

It has experimented not consciously, but through variation, selection and enormous numbers of biological generations.

The result is an extraordinary diversity of molecular mechanisms.

Some of those mechanisms may happen to operate in regimes where quantum physics becomes functionally important.

The bird does not need to "know" quantum mechanics.

The molecule does not need to "understand" a magnetic field.

The physical interaction simply occurs.

Evolution preserves arrangements that work.

That distinction is worth emphasizing because popular discussions of quantum biology sometimes go too far.

The existence of quantum effects in a biological process does not imply that the organism is consciously manipulating quantum states.

Nature does not need an engineer standing behind the mechanism.

Natural selection can discover physical solutions without understanding them.

The mystery of smell

Our sense of smell provides another intriguing case.

We are accustomed to thinking that receptors recognize molecules according to their three-dimensional shape.

The idea is intuitive.

A molecule fits into a receptor in much the same way that a key fits into a lock.

But the molecular world is more complicated than this simple analogy suggests.

Different molecules can sometimes have surprisingly similar shapes while producing different smells. Conversely, molecules with related structures can produce different sensory experiences.

This has led researchers to consider alternative mechanisms of olfaction.

One proposal, known broadly as the **vibrational theory of olfaction**, suggests that molecular vibrations may contribute to how odorant molecules are distinguished.

Quantum mechanics naturally enters such a picture because molecular vibrations are quantized.

The proposal remains an area of scientific debate. That is important.

Quantum biology does not require every quantum explanation to be accepted.

In fact, scientific disagreement is part of the field's development.

The value of such hypotheses is that they generate questions that can potentially be tested.

Does molecular shape alone explain odor discrimination?

Are vibrational properties relevant?

Can experiments distinguish competing explanations?

These are scientific questions.

And this brings us to one of the most important principles for thinking about quantum biology:

A quantum explanation is scientifically interesting only when it makes a difference that can, in principle, be tested.

Without that requirement, quantum biology can easily become a collection of attractive metaphors.

With it, quantum biology becomes science.

From strange molecules to living systems

At this point, it is tempting to take a dramatic leap.

If quantum phenomena occur in photosynthesis, enzymes, magnetoreception and perhaps olfaction, perhaps the entire organism is a quantum computer.

That would certainly make for an exciting headline.

But scientifically, we need to slow down.

There is a huge difference between saying:

Quantum mechanics contributes to biological processes.

and saying:

The organism is performing quantum computation.

The first statement can be investigated experimentally in specific systems.

The second is a much broader theoretical claim.

This distinction is especially important because the word **computation** has acquired an extremely broad meaning.

A physical system changes state according to physical laws. In one sense, information is being transformed whenever a physical state changes.

But that does not automatically make the system a useful quantum computer in the technological sense.

A modern quantum computer requires carefully controlled quantum states, operations, measurements and error management. Biological systems are enormously different.

They are warm, adaptive and continuously interacting with their environments.

The question therefore becomes much more interesting when phrased differently:

Could biological systems exploit quantum information-processing principles without resembling engineered quantum computers?

That is a question worth exploring.

And it leads directly toward the next stage of our journey.

The cell is not a quiet laboratory

There is an obvious objection to all of this.

Quantum coherence is fragile.

Biological cells are not.

A cell is crowded with molecules.

Proteins vibrate.

Water molecules collide with everything.

Ions move through channels.

Membranes fluctuate.

Chemical reactions occur continuously.

Temperature is relatively high.

The environment is noisy.

Why should delicate quantum effects survive?

This question has been central to discussions of quantum biology.

It is also where the subject becomes intellectually exciting.

The biological environment is not simply an enemy of quantum phenomena.

Sometimes the interaction with the environment is itself part of the mechanism.

A biological system is not trying to preserve a quantum state indefinitely in the way an engineered quantum computer might.

A quantum phenomenon may only need to persist long enough to influence a particular reaction or transfer process.

Milliseconds may be irrelevant.

Even extremely short timescales can matter if they alter the probability of a biologically important event.

This is an important shift in perspective.

Perhaps we have been asking the wrong question.

Instead of asking:

"How can biology maintain perfect quantum coherence?"

we might sometimes ask:

"How much quantum coherence is sufficient to influence biological function?"

That is a much more biologically meaningful question.

Living systems do not need to behave like idealized quantum laboratories.

They only need to exploit physical effects that provide an advantage.

The crowded cell and the problem of communication

This question becomes particularly interesting when we return to the cell.

In Article 2 of this series, we looked at the extraordinary organization inside a living cell.

A cell is not simply a bag containing dissolved molecules.

It is a dynamic system in which proteins, nucleic acids, membranes, organelles and signalling molecules continually interact.

Information is constantly being transmitted.

A signal arrives at the cell surface.

Receptors respond.

Molecular pathways are activated.

Transcription changes.

Proteins are produced or modified.

Metabolism shifts.

The cell changes its behaviour.

The conventional description of this process is largely biochemical.

Molecules bind.

Chemical reactions occur.

Conformations change.

Signals propagate through networks.

This framework is extraordinarily successful.

But it also raises a deeper question.

Is the classical molecular description always sufficient to explain how information is coordinated inside such a crowded system?

My earlier work explicitly raised this issue.

In my 2019 paper on quantum phenomena in biological cells, I questioned whether conventional diffusion-based descriptions could fully account for the speed and subtlety of biological communication and explored whether quantum properties of complex biomolecules might be relevant.

That was not a demonstration that non-local communication occurs inside cells.

It was a hypothesis.

That distinction matters.

The scientifically productive question is not:

"Have we proved quantum communication between biomolecules?"

We have not.

The productive question is:

Could quantum phenomena contribute to biological information processing in ways that remain experimentally undiscovered?

That question is still open.

And open questions are where science becomes interesting.

What quantum biology does—and does not—tell us

At this point, we should draw a boundary.

Quantum biology does **not** tell us that everything mysterious in biology is quantum.

It does not mean that unexplained phenomena should automatically be labelled "quantum."

It does not mean that consciousness has been shown to arise from quantum mechanics.

It does not establish telepathy.

It does not establish communication through wormholes.

It does not establish a universal consciousness field.

And it certainly does not mean that every biological molecule should be regarded as a qubit.

These are separate hypotheses, requiring separate evidence.

This distinction is particularly important for anyone working at the boundary between established science and speculative theory.

My own earlier research travelled considerably further than the experimentally established territory of quantum biology. In a series of papers, I explored ideas involving quantum computation, consciousness, biomolecular interactions, non-local communication and the possibility of a deeper relationship between consciousness and physical reality. A later paper explicitly reviewed the scientific status of these hypotheses.

Those ideas form an important part of my intellectual journey.

But they should not be confused with the established core of quantum biology.

That distinction is one I would now make much more explicitly.

There is a useful intellectual ladder here.

At the bottom is **quantum physics**, which provides the fundamental description of matter.

Above it is **quantum chemistry**, which explains molecular structure and reactions.

Then comes **quantum biology**, where quantum phenomena may contribute to biological function.

Above that is the more speculative question of **quantum information processing in biological systems**.

And beyond that lie still more ambitious questions about **consciousness and the nature of reality**.

Each step requires additional evidence.

We should not jump from the first step to the last.

Why this matters for the future of biology

Why should biologists care about any of this?

One reason is technological.

If living systems have evolved mechanisms for exploiting quantum phenomena, understanding those mechanisms could inspire new technologies.

Nature has already inspired computing architectures, materials, algorithms and engineering systems.

Perhaps biology will also teach us something about quantum information processing.

Another reason is methodological.

Modern biology increasingly operates at multiple scales.

We study atoms and molecules.

We study protein networks.

We study cells.

We study tissues.

We study organisms.

We study populations and ecosystems.

The challenge is to understand how phenomena at one level generate organization at another.

Quantum biology reminds us that the hierarchy may extend deeper than we usually consider.

The behaviour of a biological system may ultimately depend upon interactions occurring at scales far below the level at which we observe the biological function.

That does not mean that every biological problem needs a quantum explanation.

Most biological questions can and should continue to be addressed using the tools that already work.

But there may be situations in which the deeper physical description becomes essential.

Knowing when that happens may become an important part of twenty-first-century biology.

A new way of thinking about information

There is another reason quantum biology matters to this series.

It brings us back to our original subject:

information.

In Article 1, we asked whether biology might require a richer theory of information.

In Article 2, we saw that a cell is an extraordinarily organized information-processing environment.

In Article 3, we considered whether cellular sensing, integration and adaptation could reasonably be described in computational terms.

Now quantum biology adds another layer.

At the quantum level, information is not represented in exactly the same way as in a conventional digital computer.

A classical bit is either 0 or 1.

A quantum state can involve a superposition of possible states.

Quantum systems can also exhibit correlations that have no straightforward classical analogue.

This does not mean that biological information is necessarily quantum information.

But it gives us a new conceptual possibility.

Perhaps biological information is not merely a sequence of digital symbols.

Perhaps, at some physical level, biological systems exploit **states, probabilities, correlations and dynamics** that are richer than our conventional information metaphors suggest.

This possibility is one reason quantum biology is so relevant to the broader rethinking of biology.

The central question is gradually changing.

We began by asking:

What molecules make up life?

Then:

How are those molecules organized?

Then:

How does that organization process information?

Now we can ask:

What are the physical limits of biological information processing?

And that question takes us directly toward the frontier.

Could evolution have discovered something we have not yet learned to build?

Consider what technology has done during the last century.

Humans built digital computers.

Then neural networks.

Then artificial intelligence.

Now we are building quantum computers.

But biological systems have been processing information for billions of years.

The cell existed long before the transistor.

The nervous system existed long before the computer.

Evolution has had an enormous amount of time to explore physical strategies that engineers are only beginning to understand.

This does not mean that biological systems are superior to engineered computers in every respect.

They are not.

A modern computer can perform operations with extraordinary speed and precision that a cell cannot.

But biological systems possess other remarkable characteristics.

They are adaptive.

They repair themselves.

They reproduce.

They operate under uncertainty.

They continuously interact with changing environments.

They integrate information across multiple scales.

And they do all of this while operating within the constraints of ordinary physics.

Perhaps some of these capabilities arise from principles that engineering has not yet fully captured.

Quantum biology gives us one possible place to look.

Not because quantum mechanics is magical.

But because it is part of the physical reality in which life evolved.

From quantum chemistry to quantum information

This is where my own intellectual journey began to move beyond conventional quantum biology.

If quantum phenomena can influence individual biological reactions, could larger biological systems somehow exploit them collectively?

Could the organization of biomolecules matter as much as the quantum properties of individual molecules?

Could coherence, connectivity and information flow become biological variables in their own right?

These questions are much harder.

They require collaboration between biology, physics, mathematics, information theory and computation.

They also require something that science sometimes finds uncomfortable:

the willingness to say "we do not yet know."

That is not a weakness.

It is the beginning of research.

My earlier papers explored increasingly ambitious possibilities along this path. I proposed that biological organisms could be viewed from a quantum-computational perspective and considered hierarchical quantum networks extending from elementary particles through biomolecules and cells.

I also explored the possibility that complexity and coherence might provide a conceptual bridge between physical organization and biological properties.

These proposals should be understood as **hypotheses**, not established facts.

That distinction is especially important today.

The value of a hypothesis is not that we believe it.

The value is that it gives us something that can eventually be challenged.

A good scientific hypothesis should make us ask:

What would we observe if it were true?

What would we observe if it were false?

Can we design an experiment that distinguishes the two?

Can we build a mathematical model?

Can we simulate the system?

Can another researcher reproduce the result?

Those questions are ultimately more important than how fascinating the hypothesis sounds.

The danger of making quantum biology mystical

There is a temptation whenever quantum mechanics enters a discussion of consciousness, spirituality or human experience.

Quantum physics is strange.

Human consciousness is mysterious.

Therefore, perhaps the two must be connected.

That reasoning is seductive.

It is also scientifically dangerous.

The fact that quantum mechanics is counterintuitive does not make it an explanation for every phenomenon we do not understand.

A scientific theory cannot be justified simply because it sounds profound.

This is especially important when discussing spirituality.

Spiritual traditions have explored consciousness, interconnectedness, experience and the nature of reality for thousands of years. Science approaches these questions differently.

Science requires operational definitions.

It requires observations.

It requires measurements.

It requires models that can be tested against reality.

There may eventually be productive dialogue between scientific and philosophical approaches to consciousness.

But quantum mechanics should not be used as a shortcut between them.

If a connection exists, it must ultimately be demonstrated.

That is precisely why I believe quantum biology is more interesting when presented without exaggerated claims.

The real science is already fascinating enough.

A leaf, an enzyme and a bird

Let us return to the three examples with which we began.

A leaf captures light.

An enzyme accelerates a chemical reaction.

A bird senses Earth's magnetic field.

These phenomena appear completely unrelated.

One belongs to plants.

One belongs to molecular chemistry.

One belongs to animal behaviour.

Yet quantum mechanics may contribute to all three.

That is the real significance of quantum biology.

It suggests that the boundary between physics and biology may be more permeable than we once imagined.

Biology does not sit above physics as an independent layer.

It emerges from physical processes—but organizes those processes into something extraordinarily complex.

The challenge is therefore not to reduce biology to physics.

The challenge is to understand **how physical laws become biological organization**.

That is a much deeper question.

And it returns us to the central theme of this series.

Biology is increasingly becoming a science of organization and information.

Quantum biology suggests that the physical substrate on which this information is processed may sometimes have quantum characteristics.

The next question is therefore unavoidable.

From quantum effects to biological computation

Suppose, for a moment, that some biological systems can exploit quantum phenomena.

What could they do with them?

Could they improve the efficiency of energy transfer?

Could they enhance molecular recognition?

Could they provide unusual mechanisms of sensing?

Could they influence the way information is integrated within biological networks?

Could quantum correlations somehow become biologically useful at larger scales?

We do not yet know.

But notice how different this question is from the popular question:

"Are humans quantum computers?"

The first question is scientific.

The second is too broad to be useful without defining what "quantum computer" means.

A more productive approach is to start with specific biological functions.

Identify a physical phenomenon.

Measure it.

Determine whether it influences biological function.

Then ask whether the phenomenon can be integrated into a larger computational framework.

That is how we move from metaphor to science.

And that is the direction in which I would like this series to move.

The quiet revolution

Perhaps quantum biology will eventually become a major pillar of biology.

Perhaps it will remain a specialized field relevant to a limited number of biological processes.

Perhaps some of today's exciting hypotheses will be discarded.

Perhaps others will become central to our understanding of life.

Science cannot know the answer in advance.

But something important has already happened.

The old assumption that quantum phenomena are irrelevant to living systems has become increasingly difficult to maintain.

We now have biological systems in which quantum effects are seriously investigated as possible contributors to function.

That does not mean that biology has become quantum.

It means that biology has discovered that **quantum mechanics sometimes matters**.

And that is enough to open a new door.

The question that remains

We began this article with a leaf.

We saw energy moving through a molecular system.

We followed quantum tunnelling into enzyme chemistry.

We travelled with migratory animals and considered how quantum spin dynamics might contribute to magnetic sensing.

We entered the molecular world of smell.

And finally, we returned to the living cell.

What connects these stories is not a claim that biology is secretly a quantum computer.

The connection is more subtle.

Life operates in a quantum universe.

Sometimes the quantum character of that universe appears to remain buried beneath ordinary chemistry.

Sometimes, however, it may become biologically significant.

And whenever that happens, the distinction between physics and biology becomes less clear.

This raises a question that is much larger than quantum biology itself.

If living systems can exploit quantum physical phenomena, perhaps we should reconsider what we mean by **biological information**.

Information in a textbook is usually represented by letters, numbers or binary symbols.

Information in a cell is different.

It is encoded in molecular structures, concentrations, chemical modifications, spatial arrangements, timing and interactions.

And perhaps, at a deeper level, it can also involve physical states and correlations governed by quantum mechanics.

We are therefore approaching a new question:

Is biological information fundamentally different from the information processed by our machines?

If it is, then perhaps our existing theories of information capture only part of what life is doing.
That is where our journey takes us next.

From Molecules to Information

The first article in this series asked why biology might need a new information theory.

The second asked whether the cell is really just a bag of chemicals.

The third explored whether a cell can be understood as an information-processing system.

This fourth article has taken us one level deeper.

The molecules themselves live in a quantum world.

The question now is no longer simply whether quantum mechanics operates inside biology.

It clearly does.

The more interesting question is:

How might quantum physical principles shape the way biological systems represent, transmit and process information?

That is the subject of the next stage of our journey.

And it brings us to a possibility that is both exciting and difficult:

perhaps the transition from chemistry to information processing is not merely a metaphor.

Perhaps it reflects something fundamental about life.

We are not there yet.

But that is precisely why the question is worth asking.

A note on evidence and speculation

Because this series moves toward the boundary between biology, physics, information theory and consciousness, I want to make one distinction explicit.

The examples discussed here—quantum phenomena in biological chemistry, photosynthetic systems, enzyme reactions and proposed mechanisms of magnetoreception—belong to the scientific discussion of quantum biology.

My own earlier work went further, proposing broader hypotheses concerning quantum computation, biological systems, non-local communication and consciousness. Those proposals were explicitly presented as hypotheses in my earlier publications. For example, my 2019 synthesis of this work listed hypotheses ranging from quantum-computational descriptions of biological organisms to proposed quantum communication between biomolecules.

Some of those ideas remain speculative and require substantially more theoretical and experimental development.

I therefore do **not** present them here as established science.

For me, that distinction is not a retreat from the larger questions.

It is the opposite.

If we want to explore the frontier between biology, physics and consciousness seriously, we must first be clear about where the evidence ends and the hypothesis begins.

That boundary is where the real scientific adventure starts.

Rethinking Biology: A Journey from Molecules to Information

Article 1: *Why Biology Needs a New Information Theory* **Article 2:** *Is the Cell Really Just a Bag of Chemicals?* **Article 3:** *Can a Cell Compute?* **Article 4:** *The Quiet Revolution Called Quantum Biology*

Next: *From Chemistry to Information Processing*